Task:18 Configure Amazon SES SMTP with PHP and PHPMailer

Objective

To configure Amazon Simple Email Service (Amazon SES) as an SMTP mail service on a RHEL 9 AWS EC2 instance and send a test email using PHP and PHPMailer.

Requirements

- AWS EC2 instance with RHEL 9
- Amazon SES
- Verified sender email identity
- Verified recipient email identity
- SES SMTP credentials
- PHP
- Composer
- PHPMailer
- Internet connectivity from the EC2 instance

1. Configure Amazon SES

Amazon SES was configured in the **Asia Pacific (Mumbai)** – **`ap-south-1`** region.

Two email identities were created:

- Sender identity: `zenhalzanu4@gmail.com`
- Recipient identity: `adithya.pathiyadiyil@gmail.com`

The sender identity was used as the **From** address and the recipient identity was used as the **To** address.

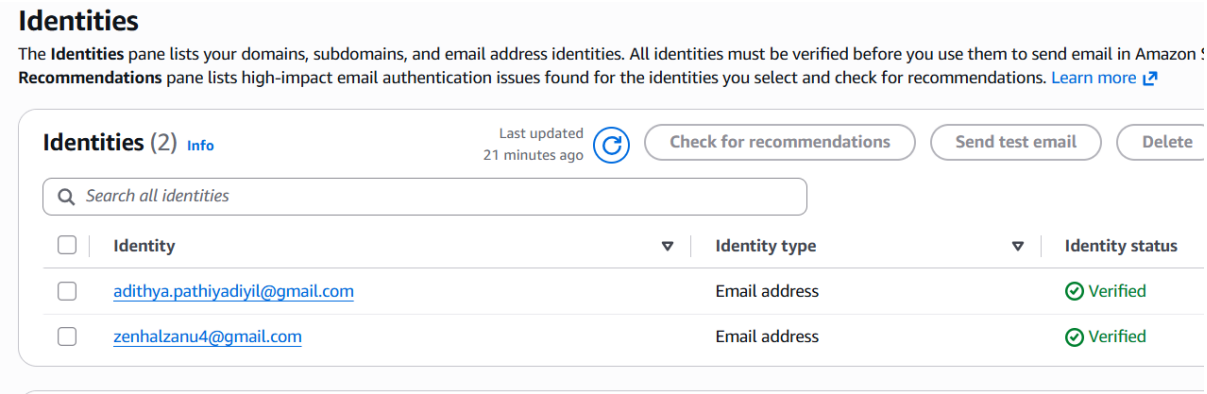
The SES SMTP endpoint used was:

```
email-smtp.ap-south-1.amazonaws.com
```

SMTP port:

```
587
```

Port 587 was used with **STARTTLS** for encrypted SMTP communication.



2. Create IAM User and SES SMTP Credentials

An
IAM user was created to allow the RHEL 9 server to authenticate with Amazon SES through SMTP.

The IAM user was created through the **Amazon SES → SMTP settings** section.

The
SMTP credentials generated for the IAM user contained:

- SMTP Username
- SMTP Password

The
SMTP credentials were used in the PHPMailer configuration for authentication.

The
credentials were configured in the PHP script as:

```
$mail->Username = 'SMTP_USERNAME';
$mail->Password = 'SMTP_PASSWORD';
```

The SES
SMTP configuration used was:

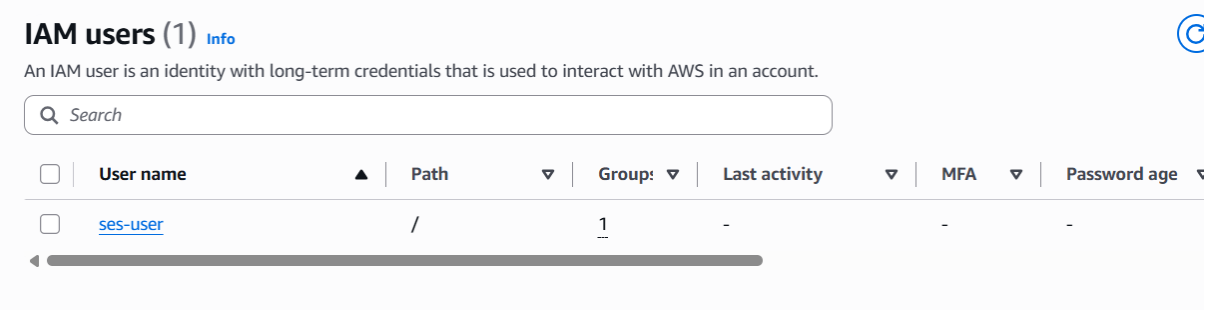
SMTP endpoint:

```
email-smtp.ap-south-1.amazonaws.com
```

SMTP
port:

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was used with **STARTTLS** for encrypted SMTP communication.



3. Connect to the RHEL 9 EC2 Instance

The RHEL 9 EC2 instance was launched and accessed through SSH.

The server was then used as the environment for configuring PHP and PHPMailer.

4. Create the PHP Project

A project directory was created:

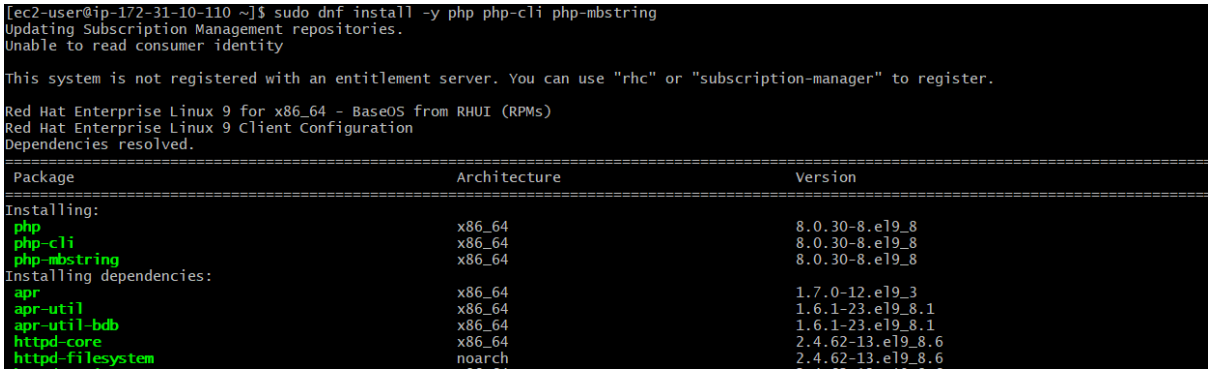
```
mkdir ses-php
cd ses-php
```

Composer was used to manage the PHP dependency.

PHPMailer was installed using:

```
composer require phpmailer/phpmailer
```

The installation completed successfully and the `vendor` directory was generated.



```
[ec2-user@ip-172-31-10-110 ~]$ php -r "copy('https://getcomposer.org/installer', 'composer-setup.php');"
[ec2-user@ip-172-31-10-110 ~]$ php composer-setup.php --check
All settings correct for using Composer
[ec2-user@ip-172-31-10-110 ~]$ sudo php composer-setup.php --install-dir=/usr/local/bin --filename=composer
All settings correct for using Composer
Downloading...

Composer (version 2.10.3) successfully installed to: /usr/local/bin/composer
Use it: php /usr/local/bin/composer

[ec2-user@ip-172-31-10-110 ~]$ composer --version
Composer version 2.10.3 2026-08-27 13:34:23
PHP version 8.0.30 (/usr/bin/php)
Run the "diagnose" command to get more detailed diagnostics output.
```

5. Configure PHPMailer

A PHP script named:

`send-email.php`

was created.

The script used PHPMailer to connect to Amazon SES through SMTP.

The important SMTP configuration was:

```
$mail->isSMTP();
$mail->Host       = 'email-smtp.ap-south-1.amazonaws.com';
$mail->SMTPAuth   = true;
$mail->SMTPSecure = PHPMailer::ENCRYPTION_STARTTLS;
$mail->Port       = 587;
```

The sender and recipient were configured as:

```
$mail->setFrom('zenhalzanu4@gmail.com', 'SES Test');
$mail->addAddress('adithya.pathiyadiyil@gmail.com');
```

The email subject and message were configured as:

```
$mail->Subject = 'Amazon SES SMTP Test';
$mail->Body    = 'This is a test email sent using Amazon SES
SMTP and PHPMailer.';
```

6. Check PHP Syntax

Before executing the script, PHP syntax was checked using:

```
php -l send-email.php
```

The result was:

```
No syntax errors detected in send-email.php
```

This confirmed that the PHP script was syntactically correct.

7. Test SMTP Connection

PHPMailer SMTP debugging was temporarily enabled:

```
$mail->SMTPDebug = 2;
```

The PHP script was executed using:

```
php send-email.php
```

The SMTP communication showed:

```
220 email-smtp.amazonaws.com ESMTP
```

The server responded to the `EHLO` command and advertised:

```
250-STARTTLS
250-AUTH PLAIN LOGIN
```

STARTTLS was successfully established.

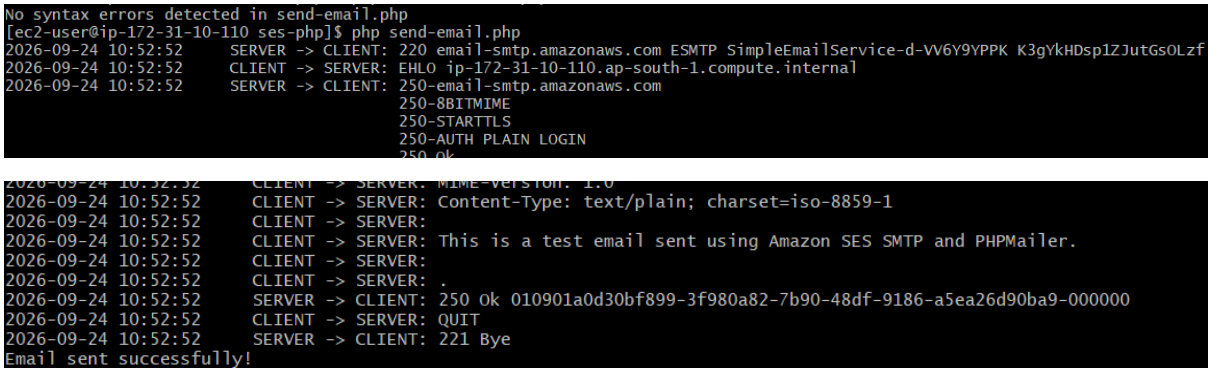
8. SMTP Authentication

PHPMailer then authenticated with the Amazon SES SMTP server.

The server returned:

```
235 Authentication successful.
```

This confirmed that the SES SMTP credentials were valid and the RHEL server could successfully authenticate with Amazon SES.



9. Send Test Email

The SMTP transaction successfully processed:

```
MAIL FROM:<sender>
250 Ok
```

and:

```
RCPT TO:<recipient>
250 Ok
```

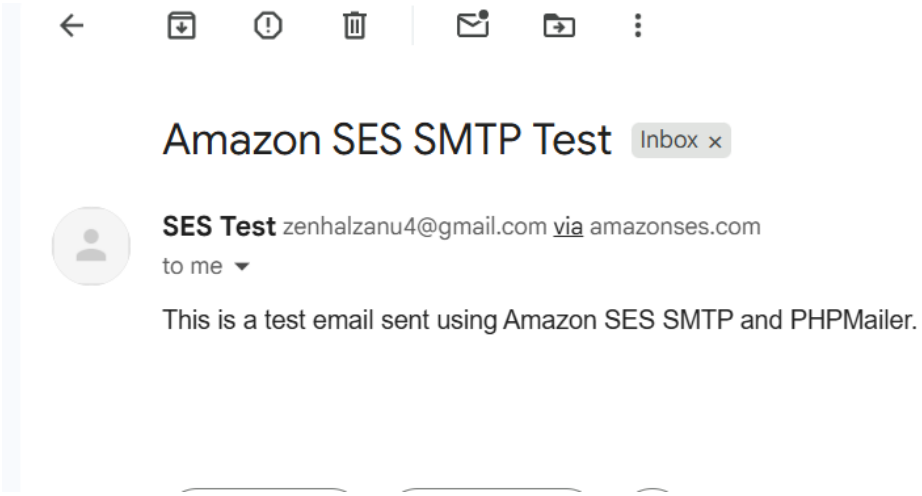
The email content was then submitted to SES.

Amazon SES returned:

```
250 Ok
```

Finally, PHPMailer displayed:

```
Email sent successfully!
```



Result

The Amazon SES SMTP configuration was successfully completed on the RHEL 9 EC2 instance.

The complete communication flow was:

```
RHEL 9 EC2
      ↓
PHP
      ↓
PHPMailer
```

```
      ↓
Amazon SES SMTP
      ↓
STARTTLS / Port 587
      ↓
Recipient Gmail
```

Final verification

- PHP installed and working
- Composer installed and working
- PHPMailer installed successfully
- SES SMTP endpoint configured
- STARTTLS connection successful
- SMTP authentication successful
- Email accepted by SES
- Test email successfully delivered to Gmail

Conclusion: Amazon SES was successfully configured as an SMTP mail service and tested using PHP and PHPMailer on a RHEL 9 AWS EC2 instance.